

RealBlend User Guide (v1.1+)

Welcome to **RealBlend** – a versatile mesh editing solution for Unity that brings professional vertex-painting, sculpting and procedural mesh creation into a single, easy-to-use tool.

I would strongly recommend following the video below to best understand how to use this asset:

<https://www.youtube.com/watch?v=qQ-P3XAelWU&t=4s>

Overview

RealBlend integrates **multi-layer vertex painting**, **intuitive sculpting tools** and **procedural mesh generation** into a single editor window. It allows you to paint up to three material layers on a mesh, sculpt geometry directly inside the Unity Editor and generate ready-to-paint floors, walls and curved surfaces. A non-destructive workflow built around the `VertexPaintStorage` component ensures that you can store vertex colours, positions and UVs without permanently modifying your original meshes.

Key features

- **Mesh sculpting tools** – push/pull geometry, slide vertices (nudge), flatten geometry along an axis, smooth surfaces, refine topology by locally subdividing triangles and remove triangles entirely. You can also revert the sculpt back to its original pose if you want to undo subtle modifications.
- **Procedural mesh creation** – generate floors, walls or curved walls with adjustable density and curvature. These meshes are ready for painting and sculpting immediately after creation.
- **Smart data storage** – RealBlend uses a `VertexPaintStorage` component to save vertex colours, positions, UVs and triangles. This allows a non-destructive workflow and makes it possible to bake the final result into a mesh asset for runtime use.
- **Mask packer utility** – pack metallic, ambient occlusion, roughness and height maps into a single four-channel texture for performance. A built-in mask packer automates the process and ensures consistency across render pipelines.
- **Shader integration** – a flexible **GBAR** channel mapping (Green/Blue/Alpha/Red) maps each vertex channel to a layer weight, wetness mask or variation mask. Adjustable shader variables such as `_BlendNoise`, `_L12_BlendBridge` and `_LayerContrast` allow you to fine-tune blending behaviour.

Set-Up

Prepare your scene

RealBlend operates directly on Unity meshes. To paint or sculpt an object:

1. **Add a MeshCollider.** Any object you wish to paint or sculpt must have a MeshCollider attached. A collider ensures the painting raycast hits the mesh accurately.
2. **Attach a VertexPaintStorage component.** This component stores vertex colours, positions, UVs and triangle data so that your edits persist. When you first select a mesh in the tool, RealBlend will prompt you to add this component automatically.
3. **Assign a compatible material.** The object must use a shader that reads vertex colours. You can use the provided RealBlendBuiltInVariation shader, included URP/HDRP Shader Graphs, or any custom shader that supports vertex colours.

Download and prepare textures

High-quality textures are essential for convincing blends. Free assets on sites such as [Polyhaven](#) and [Ambient CG](#) offer PBR-ready texture sets. Each set typically includes albedo, metallic, roughness, ambient occlusion and height maps. Download at least two or three texture sets that you intend to blend (for example: brick, concrete, moss).

Mask Packer – Combining channels into one map

Complex shaders often rely on several texture channels. To improve performance, RealBlend includes a **Mask Packer** accessible via **Tools → RealBlend → Mask Packer**. It merges multiple grayscale maps into a single four-channel texture according to the following format:

Channel	Purpose	Typical map
R (Red)	Metallic information	Metallic mask
G (Green)	Ambient Occlusion	AO map
B (Blue)	Roughness or smoothness	Roughness map (check <i>Use smoothness</i> to invert when using a smoothness texture)
A (Alpha)	Height or displacement mask	Height map

When packing masks, ensure each source texture is imported as `Single Channel` in Unity to avoid colour bleeding. Once packed, assign the resulting mask map to your material's mask slot.

Creating materials and assigning textures

1. **Create a material.** In your Unity project, right-click in the Project window and choose **Create** → **Material**. Assign the RealBlend shader variant appropriate to your render pipeline (Built-in, URP or HDRP).
2. **Load textures.** Assign your albedo/diffuse textures to the Base and Layer slots. For each layer you can specify normal maps, mask maps (created with the Mask Packer) and additional albedo textures for the variation set (Set B). Variation textures allow you to paint subtle differences (e.g., faded bricks, greener moss) without using extra materials.
3. **Tune shader variables.** The RealBlend shaders expose several variables to control the blending:
4. **_BlendNoise** – adds noise to blend edges to reduce procedural-looking transitions.
5. **_L12_BlendBridge** – smooths the transition between Layer 1 and Layer 2.
6. **_LayerContrast** – adjusts the hardness of height-based blending.
7. **Variation Set B controls (_Tint_B, _Saturation_B, etc.)** – used in combination with the Red vertex channel to control colour tint and saturation differences between the A and B variations.

Tip: Set the colours for Variation B to an obvious tint (such as green) while you fine-tune the shader. This makes it easy to see where Variation B appears during painting. Later you can adjust the tint and saturation to more subtle values.

Creating paint-ready meshes

RealBlend's **Create** tab generates simple meshes directly inside Unity and automatically attaches the required components. The following parameters let you customise the shape and density of generated meshes:

Setting	Description
Orientation	Choose Floor (XZ) , Wall (XY) or Curved Wall . Floors lie on the XZ plane; walls stand upright; curved walls bend around a circle segment.
Curvature (curved wall)	Controls how tightly the mesh curves.
Completeness (curved wall)	The angular extent of the curved wall.

Setting	Description
Pivot location	Select Mesh Centre or Circle Centre to determine the pivot for curved walls.
Density (Verts/m)	Adjusts vertex density per metre. Higher values provide more vertices for detailed painting and sculpting at the cost of a higher triangle count.

New in v1.1: A **Smart Preview** mode lets you view large procedurally generated meshes without editor slowdown. Toggle this in the Create tab when working with high vertex counts to avoid lag (no citation; feature described in the v1.1 update).

After configuring the settings, click **Create** to generate the mesh. RealBlend automatically adds a MeshCollider, attaches a VertexPaintStorage component and assigns your current material so that you can start painting immediately.

Painting workflow

RealBlend's **Paint** tab is the heart of texture blending. It offers two primary workflow modes:

1. **LayeredBlend** – paint up to three texture layers using the RGBA vertex channels and base layer. This mode is ideal for blending materials such as bricks, concrete, and moss on a single mesh.
2. **VertexColor** – paint pure vertex colours, with support for custom colour palettes (introduced in v1.1). This mode is useful for creating masks for advanced shaders or applying simple colour tints.

Brush controls

The brush settings govern how paint is applied:

- **Size** – diameter of the brush in world units.
- **Opacity** – strength of the paint; lower values produce translucent strokes.
- **Hardness** – softness of the brush edge.
- **Mouse sensitivity** – controls how quickly paint is applied as you move the mouse. *In v1.1 this sensitivity is adjustable, making it easier to paint lightly without feather-touch control.*
- **Brush Alpha & Rotation** – assign a stamp texture as an alpha mask and rotate it to produce organic, non-repeating patterns.
- **Flood Fill** – fills the entire mesh with the selected layer or colour.
- **Lock Selection** – prevents switching objects accidentally while painting.

Hold **Shift** while painting to erase paint instead of adding it. This makes it easy to remove stray strokes or revert areas to the base layer.

Painting techniques

- **Procedural layer placement.** RealBlend includes procedural painting tools that allow you to quickly cover large areas with a chosen texture. Adjust **Coverage** (percentage of the mesh covered), **Opacity** (strength) and **Edge Softness** (fall-off around edges) to achieve organic patterns. After a procedural pass, refine the result manually with the brush to break up repetition (no citation; based on video tutorial).
- **Variation painting.** The **Variation** brush paints an overlay stored in the Red channel (Layer 4). It blends between the standard texture and its Variation B counterpart. Use this to add stains, wear, or colour differences without increasing draw calls. To remove variation, switch to the Variation brush, hold **Shift** and paint over the area to erase.
- **Wetness painting.** RealBlend's shader uses the Alpha channel (Layer 3) as a **Wetness mask**. Painting into this channel darkens the surface and increases reflection, simulating damp areas, puddles or mossy corners (feature from the video; unsourced). Drive the reflection with screen-space or probe-based reflections.
- **Five-way blend.** The included **5-Way Blend Shader** allows five different texture sets to be blended on a single mesh. Assign textures to the Base and four layers, then paint across them using the LayeredBlend mode to create richly detailed surfaces (feature referenced from the tutorial; unsourced).

Tip: When painting, start with a Flood Fill to lay down the primary material, then use the procedural tools to add randomised coverage of secondary textures. Finally, use the brush to fine-tune details, emphasising wear patterns, dirt buildup or cracks along edges.

Sculpting workflow

The **Sculpt** tab modifies the actual geometry of the mesh. Each sculpting mode operates on selected vertices within the brush radius:

Mode	Purpose
Move / Push / Pull	Displace vertices along their normals or along the view direction to create bulges, dents and cracks. Adjust the brush strength and fall-off for precise control.
Nudge	Slide vertices across the plane defined by the surface, ideal for creating subtle shifts or aligning features without altering height.

Mode	Purpose
	This mode was added in v1.1 (unsourced).
Flatten Height	Constrain vertices to a specific height along the local X, Y or Z axis. Use this to create flat plateaus, clean edges or controlled slopes.
Smooth	Relax vertices and average out noise. Apply repeatedly to remove jagged edges or unwanted bumps.
Refine (Add Detail)	Subdivide triangles locally to increase vertex density. More vertices provide finer control for painting and sculpting. Caution: This changes the mesh's topology and should be used sparingly.
Remove	Delete selected triangles and any unused vertices. Useful for cutting holes (e.g., doors or windows).
Revert Pose	Gradually morph vertices back to their original stored position. Use this to soften or undo sculpting edits without affecting paint.
Vertex Select	Select individual vertices and move them using Unity's transform handles. Added in v1.1, this feature provides precise control for fine adjustments (unsourced).

Tip: Sculpting works best when your mesh has sufficient vertex density. Use the **Refine** tool locally before sculpting a detail such as a crack or indentation.

Data management & Baking

RealBlend's non-destructive workflow relies on the `VertexPaintStorage` component. This component stores the mesh's vertex colours, positions, UVs and triangles and ensures your edits survive scene reloads. When you are satisfied with your painting and sculpting, bake the mesh to create a runtime asset:

1. **Select the object** in the hierarchy.
2. In the **VertexPaintStorage** component inspector, click **Bake to Asset (Permanent)**.

3. RealBlend creates a new mesh asset in Assets/BakedVertexPaintMeshes. After baking you can remove the VertexPaintStorage component from the GameObject.

Baking is essential for builds because runtime painting and sculpting scripts do not run outside the editor.

Mask Packer reference

The **Mask Packer** utility simplifies the process of combining grayscale textures into a single mask map. Use it whenever you prepare textures for RealBlend:

1. **Open the Mask Packer** via **Tools → RealBlend → Mask Packer**.
2. Select your individual Metallic, Ambient Occlusion, Roughness/Smoothness and Height maps.
3. Click **Pack**. The utility creates a four-channel mask with the channel assignments listed above.
4. Assign this mask to the **Mask Map** slot of your material.

For the Built-in pipeline, note that the default shader lacks parallax occlusion mapping (POM) features and may require additional tuning of the **Contrast** variable to achieve convincing height effects.

Update 1.1 Highlights

The v1.1 update introduces several quality-of-life improvements and new tools based on community feedback:

1. **Mask sensitivity control**. Adjust the brush's sensitivity so you no longer need feather-light mouse movements when applying subtle paint.
2. **Vertex colour painting module**. Paint arbitrary vertex colours and organise them into palettes to create custom masks or drive advanced shaders.
3. **Smart Preview mode**. Preview large procedural meshes in the Create tab without editor lag (especially useful for high-density meshes).
4. **Nudge sculpting mode**. Slide vertices along the surface plane to adjust geometry without altering height or adding new triangles.
5. **Flatten Height tool**. Constrain vertices to a specific height along the X, Y or Z axis, making it easier to create clean plateaus and sharp edges.
6. **Limited built-in shader for the Built-in Render Pipeline**. Although less feature-rich than the HDRP/URP shaders, this shader provides basic blending without parallax occlusion mapping (you may need to adjust the Contrast parameter for best results).
7. **Vertex Select tool**. Pick individual vertices and move them using Unity's transform handle for precise adjustments.

Best practices & tips

- **Use high-resolution textures.** Quality base images reduce repetition and hold up under close inspection.
 - **Match texel density.** Ensure your base and layer textures have similar texel densities so that blends appear consistent.
 - **Balance triangle count and draw calls.** Modern GPUs are often limited by draw calls rather than triangle counts. Increasing mesh density (with the **Refine** tool or high density in the Create tab) may slightly increase the triangle count but reduces the need for multiple materials and hence reduces draw calls.
 - **Stay organised.** Name your layers (e.g., *Brick*, *Concrete*, *Moss*, *Dirt*, *Variation*) and materials clearly. Keep a library of high-quality albedos and mask maps for reuse across projects.
 - **Experiment with Variation.** The Red channel Variation mask and Set B textures provide an inexpensive way to introduce randomness. Use it sparingly to break up uniformity.
 - **Sculpt after painting when possible.** Painting first establishes the material distribution; sculpting afterwards allows you to accentuate or hide the painted details. However, if you add triangles later (with **Refine**), the newly created vertices will inherit colours via interpolation.
 - **Bake for builds.** Always bake your painted/sculpted meshes before making builds, otherwise RealBlend will attempt to run editor-only scripts at runtime.
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